BADJI MOKHTAR UNIVERSITY ANNABA

FACULTY OF MEDICINE

FIRST YEAR OF MEDICINE

PHYSIOLOGY

CELLULAR ELECTROPHYSIOLOGY

Student Course Handout

Dr A. Benahmed

Assistant Lecturer in Clinical Neurophysiology

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Learning objectives

By the end of this lecture, you should be able to:

1. Explain how the resting membrane potential arises and why it is close to E_K.
2. Calculate an equilibrium potential with the Nernst equation and predict ion movement from the driving force.
3. Describe the phases of the action potential and the role of voltage-gated Na^+ and K^+ channels in each.
4. Explain threshold, the all-or-none law and the refractory periods.
5. Compare continuous and saltatory conduction and the factors that set conduction velocity.
6. Distinguish graded potentials from action potentials and explain summation.

Conventions used in all diagrams

Na^+ in red · K^+ in purple · ↑ = movement **out** of the cell · ↓ = movement **into** the cell

Red box = active (firing) membrane · yellow = membrane being depolarized by local current · blue-grey = refractory membrane · green = myelin

Voltages are illustrative values for a typical neuron (rest ≈ −70 mV, threshold ≈ −55 mV); they vary with cell type and conditions.

1. Introduction

Cellular electrophysiology studies the electrical properties of cells and tissues: how electrical signals are **generated**, **transmitted** and **recorded**. These signals underlie neuronal communication, sensory perception, muscle contraction, the heart rhythm and neurosecretion.

Why it matters in medicine

System	Electrical function	Disorder when it fails	Clinical recording
Brain	Neuronal communication	Epilepsy	Electroencephalography (EEG)
Peripheral nerve and muscle	Conduction and contraction	Neuropathies, myopathies	Electroneuromyography (ENMG): EMG and nerve conduction studies
Heart	Rhythm and coordinated contraction	Arrhythmias	Electrocardiography (ECG)

Electrical treatments also rely on these principles: a **pacemaker** delivers impulses to support the heart rhythm, and **cardiac ablation** destroys tissue that generates or conducts abnormal rhythms.

Excitable cells

An **excitable cell** responds to a sufficient stimulus by changing its membrane potential and can generate **action potentials**. Excitability depends on ion gradients and on ion channels in the plasma membrane.

Diagram 1. The four excitable cell types and the common sequence

Neuron
Action potentials carry information.

Skeletal muscle fibre
Action potentials trigger contraction.

Cardiac muscle cell
Action potentials coordinate the heartbeat.

Smooth muscle cell
Many generate action potentials; behaviour varies between tissues.

Sufficient stimulus → **Membrane depolarization** → **Action potential** → **Cellular response** signal, contraction, secretion

2. Neuron electrophysiology

A neuron **receives**, **integrates** and **transmits** information. Think of it as a circuit: dendrites are the input, the soma is the processor and the axon is the output cable. Information generally flows from the dendrites and soma towards the axon terminals.

Diagram 2. Functional zones of a neuron

Dendrites and soma
input and integration
graded potentials
→
Axon hillock and initial segment
trigger zone: many Na_V channels
AP initiated
→
Axon
conduction
AP propagated
→
Axon terminals
output
transmitter release

Part	Structure	Function
Soma (cell body)	Nucleus, mitochondria, organelles; Nissl substance (rough ER and ribosomes)	Metabolism and protein synthesis (enzymes, channels, peptide precursors); integrates synaptic inputs
Dendrites	Branching processes; dendritic spines increase synaptic surface	Receive neurotransmitter signals and produce local voltage changes
Axon	Starts at the axon hillock; microscopic to over 1 m long; branches into terminals	Carries action potentials away from the soma
Myelin sheath	Lipid-rich wrapping: Schwann cells (PNS), oligodendrocytes (CNS)	Reduces current loss across the internode; speeds conduction
Nodes of Ranvier	Gaps between myelin segments, rich in Na_V channels	Sites where the action potential is regenerated (saltatory conduction)
Synapse	Junction between a neuron and another cell	Transmits the signal, usually chemically

Clinical link

In **demyelinating neuropathies**, loss of myelin slows conduction and can cause **conduction block**. Nerve conduction studies show this as reduced conduction velocity, prolonged latencies or a drop in response amplitude across the affected segment.

3. Electrical properties of excitable membranes

- **Polarization**: a voltage difference exists across the membrane; at rest the inside is negative relative to the outside.
- **Excitability**: the ability to change membrane potential in response to a stimulus (graded potential, action potential, contraction or secretion).
- **Conduction**: the ability to propagate an electrical signal along the membrane.

Types of membrane potential

Potential	Main feature	Example
Resting membrane potential	Stable baseline voltage of a cell at rest	Neuron ≈ −70 mV
Graded potential	Local change; amplitude varies with the stimulus	EPSP, IPSP, receptor potential
Action potential	Rapid, regenerative, all-or-none, propagated	Axon, skeletal muscle
Pacemaker potential	Spontaneous depolarization towards threshold	Sinoatrial node
Slow waves	Slow rhythmic voltage fluctuations	Gastrointestinal smooth muscle

4. Resting membrane potential

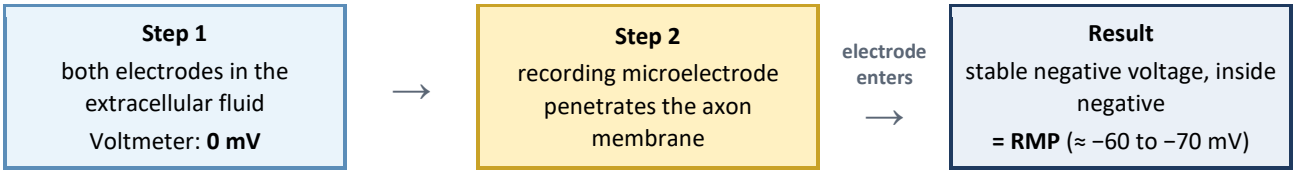
The **resting membrane potential (RMP)** is the voltage inside a cell relative to the extracellular fluid when the cell is at rest. It is expressed in millivolts: −70 mV means the inside is 70 mV negative relative to the outside. **All cells** have a membrane potential, including cells that never fire action potentials.

Category	Cell type	Approximate RMP
Excitable	Neuron	≈ −70 mV
	Skeletal muscle fibre	≈ −85 to −90 mV
	Ventricular muscle cell	≈ −85 to −90 mV
Non-excitable	Red blood cell	≈ −10 mV
	Epithelial cell, fibroblast, adipocyte	≈ −20 to −50 mV (very variable)

Excitable cells usually have more negative values, but a negative RMP alone does not make a cell excitable: the types of membrane channel matter. Cardiac **pacemaker cells** have no stable resting potential.

Demonstrating the resting potential

Diagram 3. Recording the RMP in a squid giant axon



Origin of the resting potential

The RMP arises from **unequal ion concentrations** across the membrane together with **selective permeability**. A concentration difference alone creates no voltage unless the membrane lets one ion cross more easily than the others. K^+ is concentrated inside; Na^+, Cl^- and free Ca^{2+} are concentrated outside.

Ion (squid giant axon)	Inside (mM)	Outside (mM)
K^+	400	20
Na^+	50	440
Cl^-	40–150	560
Free Ca^{2+}	0.0001	10
Impermeant anions (proteins$^-$)	High	Low

Classic squid values from the lecture; do not use them for human cells (see the mammalian table below).

Chemical and electrical forces

- **Chemical force (concentration gradient)**: moves an ion from high to low concentration.
- **Electrical force**: depends on membrane voltage and ion charge. A negative interior attracts cations and repels anions.
- Together they form the **electrochemical gradient**. Actual movement also requires an open channel for that ion.

Diagram 4. How the K^+ equilibrium potential develops (membrane permeable only to K^+)

1. Chemical force dominates
K^+ ↑ leaves down its gradient
V_m = 0, net efflux

→

2. Charge separates
inside becomes negative
electrical force pulls K^+ back ↓

→

3. Equilibrium
chemical = electrical force
no net K^+ movement
$V_m = E_K \approx -90$ mV

At equilibrium the concentration gradient still exists and ions still cross in both directions, at equal rates. Very few ions need to move to create this voltage.

Nernst equation

The **equilibrium potential** of an ion (E_{ion}) is the voltage at which its chemical and electrical forces are equal and opposite. It is calculated with the Nernst equation:

$$E_{ion} = (RT / zF) \cdot \ln ([ion]_{out} / [ion]_{in})$$

R = gas constant · T = absolute temperature (K) · z = signed charge of the ion · F = Faraday constant

At 37 °C: $E_{ion} \approx (61.5 / z) \cdot \log_{10} ([ion]_{out} / [ion]_{in})$ mV

Worked example: K^+ at 37 °C

$E_K = 61.5 \times \log_{10}(4.8 / 140) = 61.5 \times (-1.46) \approx$ **−90 mV**.

Do not forget z: for Cl^-, z = −1, which reverses the sign; for Ca^{2+}, z = +2, which halves the factor (≈ 30.8 mV).

Ion (mammalian)	Inside (mM)	Outside (mM)	E_{ion} (37 °C)
K^+	140	4.8	≈ −90 mV
Na^+	15	145	≈ +61 mV
Cl^-	10	120	≈ −66 mV
Free Ca^{2+}	0.0001	1.2	≈ +125 mV

Ionic driving force

The **driving force** on an ion is $V_m - E_{ion}$. When channels for that ion are open, the current is:

$$I_{ion} = g_{ion} \cdot (V_m - E_{ion})$$

g = conductance (how many channels are open) · positive current = outward movement of positive charge

Rule for cations: positive driving force → ion leaves; negative → ion enters. For anions (Cl^-) the ion moves the opposite way to the current.

Diagram 5. Forces acting on each ion at V_m = −70 mV

Ion	E_{ion}	Chemical force	Electrical force (inside −)	$V_m - E_{ion}$	Net movement
K^+	−90 mV	↑ out	↓ in	+20 mV	↑ **out** (moderate)
Na^+	+61 mV	↓ in	↓ in	−131 mV	↓ **in** (strong)
Cl^-	−66 mV	↓ in	↑ out (repelled)	−4 mV	↑ **out** (weak)

Ion	E_{ion}	Chemical force	Electrical force (inside −)	$V_m - E_{ion}$	Net movement
Ca^{2+}	+125 mV	↓ in	↓ in	−195 mV	**↓ in** (very strong)

These are predicted movements if the channels are open. At $V_m = E_{ion}$, there is no net movement of that ion.

Why do these ionic differences exist?

Diagram 6. Three mechanisms behind the resting potential

1. Na^+/K^+-ATPase pump
Uses ATP to move 3 Na^+ out and 2 K^+ in. Creates and maintains the gradients.

2. Selective permeability
At rest the membrane is much more permeable to K^+ than to Na^+ ($P_K \gg P_{Na}$).

3. Impermeant intracellular anions
Proteins and other trapped anions shape ion distribution (Gibbs–Donnan effect).

↓ ↓ ↓

Resting membrane potential: a small charge separation across the membrane; inside negative.
Bulk fluids on each side remain almost electrically neutral.

Diagram 7. The resting membrane: channels, pump and trapped anions

Extracellular	K^+ ↑ leaves (large flux)	Na^+ ↓ enters (small flux)	no flow	↑ 3 Na^+ out	
Membrane	K^+ leak channels: many	Na^+ leak: few	Na_V channels: closed, available	Na^+/K^+-ATPase	—
Intracellular				↓ 2 K^+ in · ATP → ADP + Pi	A^- (proteins): cannot cross

Each pump cycle moves one net positive charge out, so the pump is electrogenic and directly adds a few millivolts of negativity.

Selective permeability: why the RMP is close to E_K

Because K^+ leak conductance dominates at rest, the RMP lies close to E_K (−90 mV). The small Na^+ leak pulls it slightly towards E_{Na}, giving ≈ −70 mV. The **Goldman–Hodgkin–Katz (GHK) equation** expresses this weighting by permeability:

$$V_m = 61.5 \cdot \log_{10} \left(P_K[K^+]_{out} + P_{Na}[Na^+]_{out} + P_{Cl}[Cl^-]_{in} \right) / \left(P_K[K^+]_{in} + P_{Na}[Na^+]_{in} + P_{Cl}[Cl^-]_{out} \right)$$

Example with $P_K : P_{Na} = 1 : 0.05$ and the mammalian values above (Cl^- ignored): $V_m \approx$ **−66 mV**

The ion with the highest permeability pulls V_m towards its own equilibrium potential.

Impermeant intracellular anions (proteins, nucleic acids, organic phosphates) cannot leave the cell. They influence the distribution of permeant ions (Gibbs–Donnan principle). Only a tiny excess of charge next to the membrane is needed to produce the membrane voltage.

Clinical link: plasma potassium

Hyperkalaemia makes E_K less negative, so the RMP depolarizes. Prolonged depolarization inactivates Na_V channels and reduces excitability: this underlies the danger of arrhythmias and muscle weakness.

Hypokalaemia makes E_K more negative and hyperpolarizes the membrane, which also impairs muscle and cardiac function.

Key points

RMP = ion gradients (maintained by the pump) + selective permeability (K^+ dominates).

Each ion is pulled towards its own E_{ion}; the membrane settles where the permeability-weighted balance lies (GHK).

Driving force = $V_m - E_{ion}$; current = g × driving force.

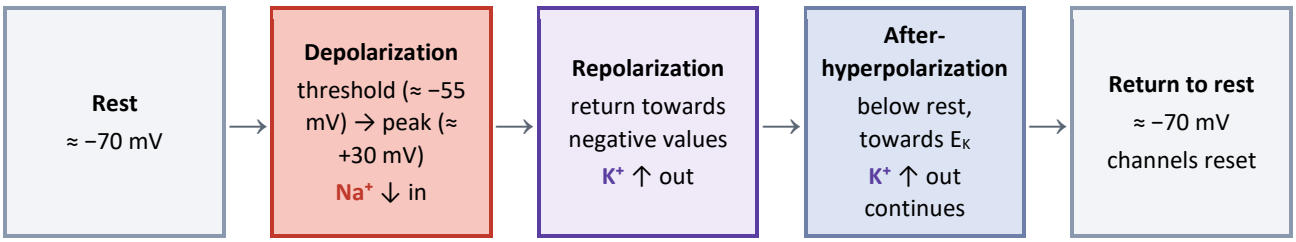
5. The action potential

An **action potential (AP)** is a rapid, transient, regenerative reversal of the membrane potential in an excitable cell. It is triggered when depolarization reaches **threshold**, it is **all-or-none**, and it is the basic signal for communication and contraction.

Experimental demonstration (squid axon): two electrodes stimulate and two record. A subthreshold stimulus gives only a small local response. A suprathreshold stimulus gives a full AP of constant amplitude and duration; a stronger stimulus does not make it bigger.

Phases and their ionic basis

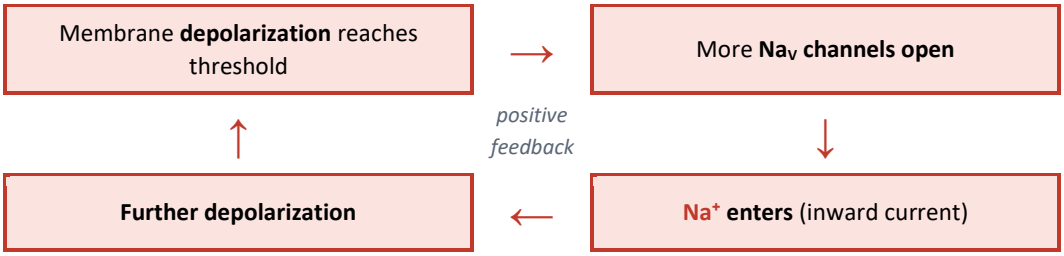
Diagram 8. The phases of a neuronal action potential



Phase	Voltage	Na_V channels	K_V channels	Dominant current
Rest	≈ −70 mV	Closed, available	Closed (leak K^+ channels open)	Small K^+ leak out
Depolarization	−55 → ≈ +30 mV	**Open** (fast activation)	Mostly still closed	**Na^+ in**
Repolarization	+30 → −70 mV	**Inactivated**	**Open** (delayed)	**K^+ out**
Afterhyperpolarization	below −70 mV	Recovering (closed)	Still open, closing slowly	K^+ out
Return to rest	≈ −70 mV	Closed, available	Closed	Leak; pump maintains gradients

During the rising phase V_m moves towards $\mathbf{E_{Na}}$; during the falling phase it moves towards $\mathbf{E_K}$. The AP therefore reflects a change in permeability, first to Na^+, then to K^+. The two currents overlap in time.

Diagram 9. The regenerative (positive) feedback of the rising phase



The loop is stopped by Na_V inactivation and by the delayed opening of K_V channels.

Voltage clamp: separating the ionic currents

The **voltage clamp** holds membrane voltage at a chosen value. The current the amplifier injects to keep it there equals, and is opposite to, the ionic current crossing the membrane, so that current can be measured. Convention: inward current (Na^+ entry) is drawn **downwards** (negative); outward current (K^+ exit) is drawn **upwards** (positive).

Diagram 10. Voltage step to 0 mV: currents recorded with and without blockers

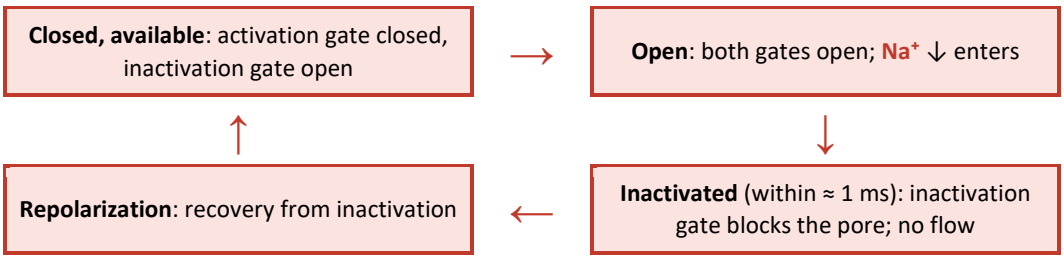
Condition	Early current (first ms)	Late current	Conclusion
Control	↓ transient **inward**	↑ sustained **outward**	Two separate currents
+ TTX (blocks Na_V)	absent	↑ outward remains	Late current = **K^+** (I_K)
+ TEA (blocks K_V)	↓ inward remains	absent	Early current = **Na^+** (I_{Na})

Conductance	Time course	Effect on the AP
g_{Na}	Rises very fast, then falls by inactivation even if depolarization persists	Rising phase
g_K (delayed rectifier)	Rises more slowly and stays high longer	Falling phase and afterhyperpolarization

Patch clamp: a glass micropipette forms a high-resistance seal on a small membrane patch. It can record a single channel, seen as small square current steps (open or closed), or the whole cell. The sum of many single-channel openings produces the macroscopic currents.

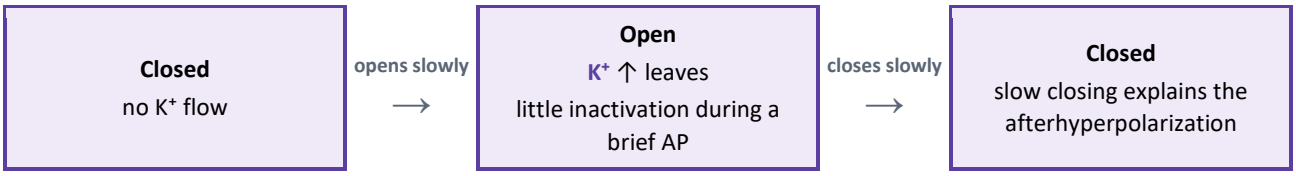
Voltage-gated channels

Diagram 11. States of the voltage-gated Na^+ channel (Na_V)



An inactivated channel cannot reopen until the membrane repolarizes: this is the basis of the absolute refractory period.

Diagram 12. The voltage-gated K^+ channel (K_V, delayed rectifier)



Feature	Na_V channel	K_V channel
Structure	One α subunit with **4 homologous domains** (I–IV)	**4 separate α subunits** around a central pore
Segments	Each domain: 6 transmembrane segments (S1–S6)	Each subunit: 6 segments (S1–S6)
Voltage sensor	S4 of each domain	S4 of each subunit
Pore	S5–S6 loops of the four domains	S5–S6 region with selectivity filter
Gating	Fast activation, then fast inactivation	Delayed activation, little fast inactivation
Current	Inward, brief	Outward, slower, sustained

Threshold and the all-or-none law

Threshold (≈ −55 mV in many neurons) is the level of depolarization at which the inward Na^+ current becomes self-sustaining. Above it, the AP has a stereotyped amplitude and shape in a given axon.

Stimulus	Response	Comment
Subthreshold	Local graded potential that fades	Size varies with the stimulus; no AP
Threshold	Full action potential	Regenerative Na^+ entry starts
Suprathreshold	Same action potential	A stronger or longer stimulus raises **firing frequency**, not AP size

Refractory periods

Diagram 13. Excitability after an action potential (timeline)

Absolute refractory period	Relative refractory period	Normal excitability
From threshold to late repolarization (≈ 1 ms). No AP possible.	Late repolarization and afterhyperpolarization (a few ms). A stronger stimulus is needed.	Back at rest.

	Absolute refractory period	Relative refractory period
Cause	Na_V channels open or inactivated	Some Na_V channels still inactivated; K^+ conductance still high (membrane hyperpolarized)
Stimulus needed	None works, however strong	Stronger than normal
Consequences	Limits maximum firing frequency	Keeps propagation one-way along the axon

Clinical and pharmacological links

Local anaesthetics (e.g. lidocaine) block Na_V channels and stop conduction in the nerve.

Tetrodotoxin (TTX), from puffer fish, blocks Na_V channels; poisoning causes paralysis.

6. Conduction of the action potential

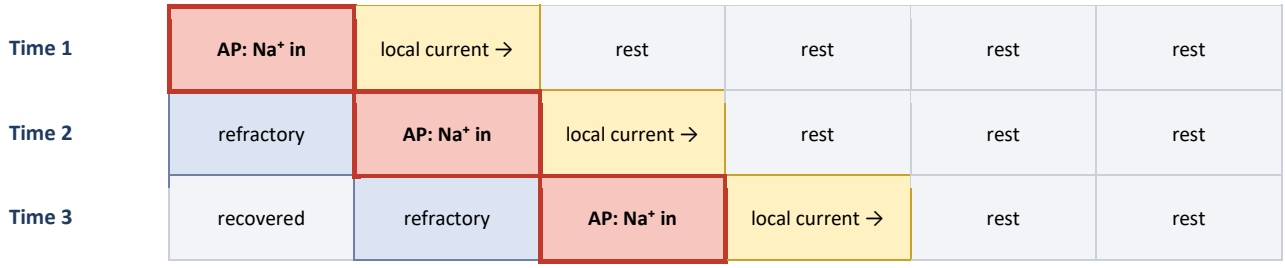
Demonstration: a squid axon is stimulated at one point and recorded at increasing distances. The AP appears progressively **later** (latency increases) but its **amplitude stays the same**: it is regenerated along the membrane instead of fading like a passive signal.

> **Conduction velocity = distance / difference in latency**
>
> Example: recording sites 10 mm and 30 mm from the stimulus, latencies 0.5 ms and 1.5 ms → v = 20 mm / 1.0 ms = **20 m/s**

Ionic mechanism of propagation

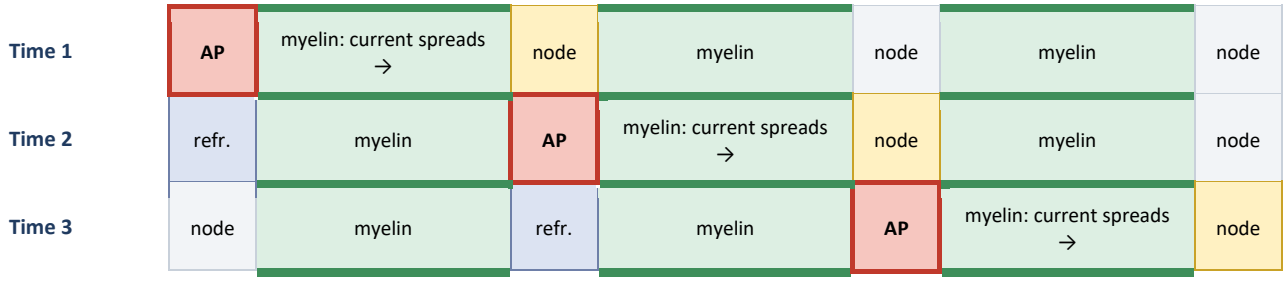
Positive charge entering the active region spreads inside the axon (**local current**) and depolarizes the next segment. When that segment reaches threshold, its Na_V channels open and the AP is regenerated. The segment just behind is **refractory**, so the impulse moves forward only. (A stimulus applied in the middle of an axon sends APs in both directions.)

Diagram 14. Continuous conduction in an unmyelinated axon



Direction of propagation → (the AP is regenerated in every successive segment)

Diagram 15. Saltatory conduction in a myelinated axon



The AP is regenerated only at the nodes of Ranvier: it "jumps" from node to node

	Unmyelinated axon	Myelinated axon
Mode	Continuous: every patch of membrane regenerates the AP	Saltatory: regenerated at the nodes only
Typical velocity	≈ 0.5–2 m/s	≈ 2–120 m/s
Energy cost	Higher: ion exchange along the whole membrane	Lower: ion exchange limited to the nodes
Example	C fibres (slow pain, autonomic)	Motor fibres and large sensory fibres

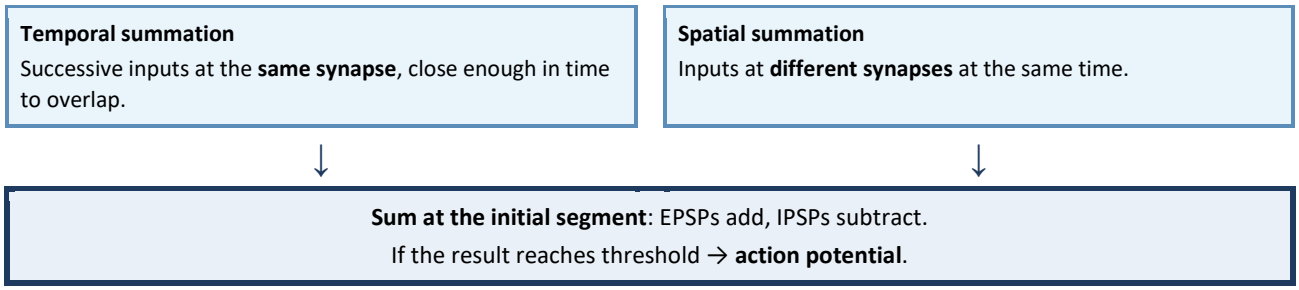
Factors that increase conduction velocity: myelination, larger axon diameter (lower internal resistance) and higher temperature (within physiological limits).

7. Graded potentials

A **graded potential** is a **local** change in membrane potential whose amplitude depends on the strength and duration of the stimulus. It can depolarize or hyperpolarize, it spreads passively and **decreases with distance**, and it is not all-or-none. Graded potentials occur in dendrites, the soma, sensory receptors and at the motor end plate.

Summation

Diagram 16. Temporal and spatial summation at the trigger zone



Types of graded potential

Type	Effect	Where
EPSP (excitatory postsynaptic potential)	Usually depolarizing; makes an AP **more** likely	Postsynaptic membrane
IPSP (inhibitory postsynaptic potential)	Hyperpolarizing or shunting; makes an AP **less** likely	Postsynaptic membrane
Receptor (generator) potential	Amplitude proportional to stimulus intensity	Sensory receptors
Pacemaker potential	Slow spontaneous depolarization that triggers APs	Sinoatrial node, some smooth muscle

These potentials are covered in detail in later chapters (synapse, sensory receptors, heart).

Graded potential versus action potential

Feature	Graded potential	Action potential
Amplitude	Variable, proportional to the stimulus	Stereotyped (all-or-none)
Spread	Passive, decreases with distance	Regenerated, constant amplitude
Summation	Yes, temporal and spatial	No
Threshold	None	Required
Refractory period	None	Yes
Direction	Depolarizing or hyperpolarizing	Always depolarizing
Main channels	Ligand-gated or mechanically gated	Voltage-gated Na^+ and K^+

8. Key takeaways

- The **RMP** depends on ion gradients (maintained by the Na^+/K^+ pump), selective permeability (K^+ dominates) and impermeant intracellular anions.
- The **Nernst equation** gives the equilibrium potential of one ion; the **GHK equation** gives the membrane potential for several permeant ions.
- Rapid Na_V activation drives the **rising phase**; Na_V inactivation and delayed K_V opening drive **repolarization** and afterhyperpolarization.
- The AP is **all-or-none**; stimulus strength is coded by firing frequency.
- **Refractory periods** limit firing frequency and keep conduction one-way.
- Conduction uses **local currents**; myelin (saltatory conduction) and larger diameter increase velocity.
- **Graded potentials** are local, variable and summable; they decide whether an AP starts.

Next chapters: synaptic transmission, sensory receptors and cardiac electrophysiology.

Abbreviations

Abbr.	Meaning	Abbr.	Meaning
RMP	Resting membrane potential	**V_m**	Membrane potential
AP	Action potential	**E_{ion}**	Equilibrium potential of an ion
Na_V / K_V	Voltage-gated Na^+ / K^+ channel	**g_{ion}**	Conductance for an ion
EPSP / IPSP	Excitatory / inhibitory postsynaptic potential	**GHK**	Goldman–Hodgkin–Katz
TTX	Tetrodotoxin (Na_V blocker)	**TEA**	Tetraethylammonium (K^+ channel blocker)

Cellular Electrophysiology

English–Arabic terminology | First-year medical physiology

Arabic equivalents are provided for the meaning used in this course.

Key terms

English	العربية
Electrophysiology	الفيزيولوجيا الكهربائية
Excitable cell	خلية قابلة للاستثارة
Neuron	عصبون (خلية عصبية)
Skeletal muscle fiber	ليف عضلي هيكلي
Cardiac muscle cell	خلية عضلية قلبية
Cell body / soma	جسم الخلية العصبية
Dendrite	تغصّن عصبي
Axon	محور عصبي
Axon hillock	تلّ المحور العصبي
Axon initial segment	القطعة الابتدائية للمحور
Axon terminal	النهاية المحورية
Synapse	المشبك العصبي
Myelin sheath	غمد الميالين
Schwann cell	خلية شوان
Oligodendrocyte	خلية قليلة التغصّن
Node of Ranvier	عقدة رانفييه
Membrane potential	جهد الغشاء
Resting membrane potential	جهد الغشاء في حالة الراحة
Polarization	الاستقطاب
Depolarization	إزالة الاستقطاب
Repolarization	إعادة الاستقطاب
Hyperpolarization	فرط الاستقطاب
Excitability	قابلية الاستثارة
Conduction	التوصيل
Stimulus	مُنَبِّه
Threshold	عتبة الاستثارة
Action potential	جهد الفعل
Graded potential	جهد متدرّج
Pacemaker potential	جهد الناظمة
Slow wave	موجة بطيئة
Ion concentration	تركيز الأيون
Concentration gradient	تدرّج التركيز
Electrical force	القوة الكهربائية
Electrochemical gradient	التدرّج الكهروكيميائي
Ion permeability	نفاذية الأيون
Equilibrium potential	جهد التوازن
Potassium equilibrium potential	جهد توازن البوتاسيوم
Nernst equation	معادلة نرنست
Driving force	القوة الدافعة
Ionic current	التيار الأيوني
Ion conductance	الناقلية الأيونية

Sodium–potassium ATPase	إنزيم مضخة الصوديوم والبوتاسيوم
Leak channel	قناة تسرّب
Voltage-gated sodium channel	قناة صوديوم ذات بوابة جهدية
Voltage-gated potassium channel	قناة بوتاسيوم ذات بوابة جهدية
Activation	التنشيط
Inactivation	التعطيل
All-or-none law	قانون الكل أو لا شيء
Absolute refractory period	فترة الجموح المطلقة
Relative refractory period	فترة الجموح النسبية
Propagation	الانتشار على طول الغشاء
Local current	تيار موضعي
Conduction velocity	سرعة التوصيل
Saltatory conduction	التوصيل القفزي
Continuous conduction	التوصيل المستمر
Voltage clamp	تثبيت الجهد
Patch clamp	تثبيت الرقعة الغشائية
Tetrodotoxin	تيترودوتوكسين
Tetraethylammonium	رباعي إيثيل الأمونيوم
Excitatory postsynaptic potential	جهد ما بعد مشبكي استثاري
Inhibitory postsynaptic potential	جهد ما بعد مشبكي تثبيطي
Spatial summation	الجمع المكاني
Temporal summation	الجمع الزماني
Electrocardiography (ECG)	تخطيط كهربائية القلب
Electroencephalography (EEG)	تخطيط كهربائية الدماغ
Electromyography (EMG)	تخطيط كهربائية العضلات

Useful verbs

English	العربية
To stimulate	ينبّه
To excite	يستثير
To depolarize	يزيل الاستقطاب
To repolarize	يعيد الاستقطاب
To hyperpolarize	يُحدث فرط الاستقطاب
To generate	يولّد
To conduct	يوصّل
To propagate	ينتشر أو ينقل
To reach threshold	يبلغ عتبة الاستثارة
To open	يفتح
To close	يغلق
To inactivate	يعطّل
To recover	يستعيد الحالة السابقة
To record	يسجّل
To measure	يقيس
To trigger	يُطلِق أو يُحفّز
To sum	يجمع